Assemble the left-hand HandUMI data collector shown in the CAD views. Use the released left_handumi STL/STEP files and the bill of materials (BOM) to identify printed parts, electronics, fasteners and quantities. Verify fastener specifications and fits against the controlled assembly drawing.

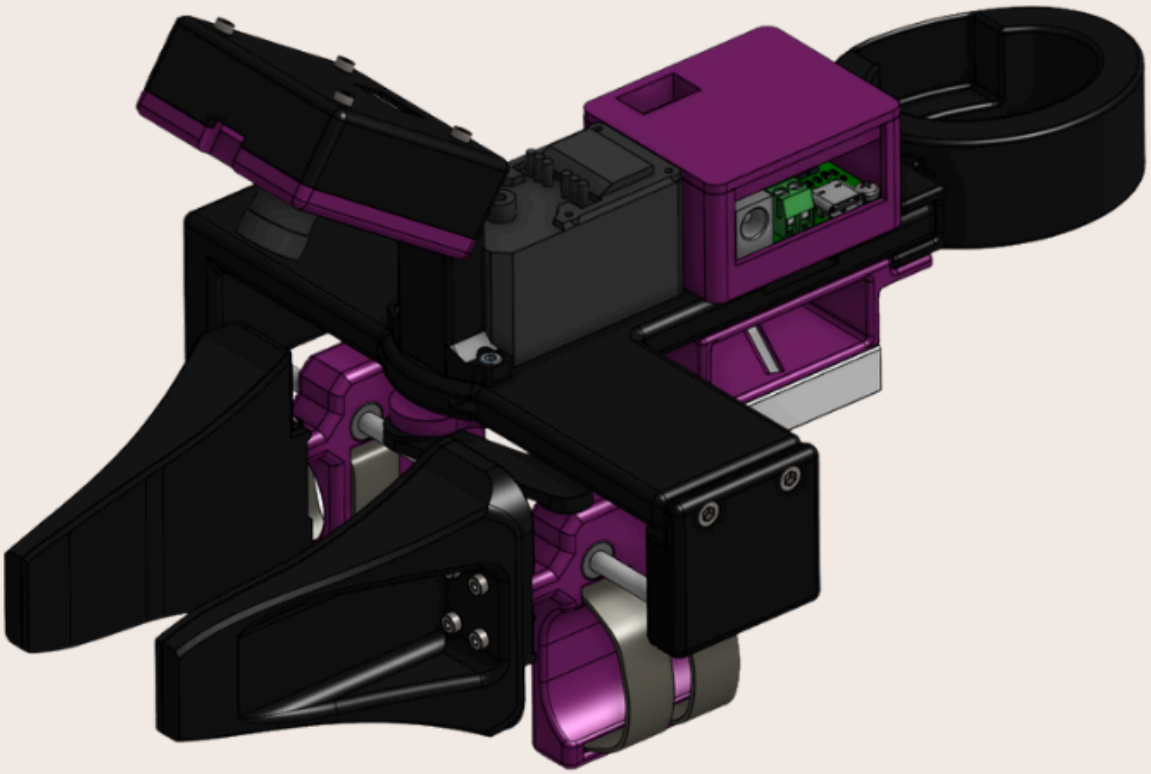


Figure 1. Completed assembly reference.

Assembly Procedure

01. Prepare the Structural Frame

Identify the hand_support_base and inspect its guide slot, mounting holes and mating surfaces. Remove any print residue that could obstruct assembly. See Figure 2.

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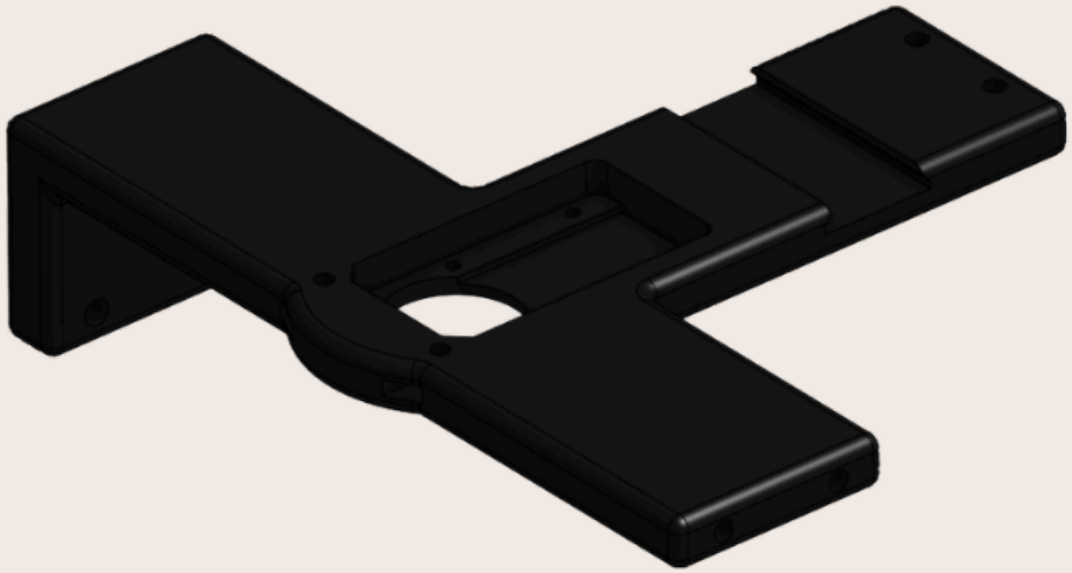


Figure 2. Hand support base before component installation.

02. Install the Hand and Controller Supports

Align the left_controller_support with the guide slot in the HandUMI main support (handumi-main-support) and slide it into position. Bond the hand_support_foam_pad to the hand_support_base at the location shown, then position the padded base beneath the controller support. Install the M3 × 16 socket-head cap screw and M3 hex nut in the indicated joint. Confirm that the pad and support remain aligned.

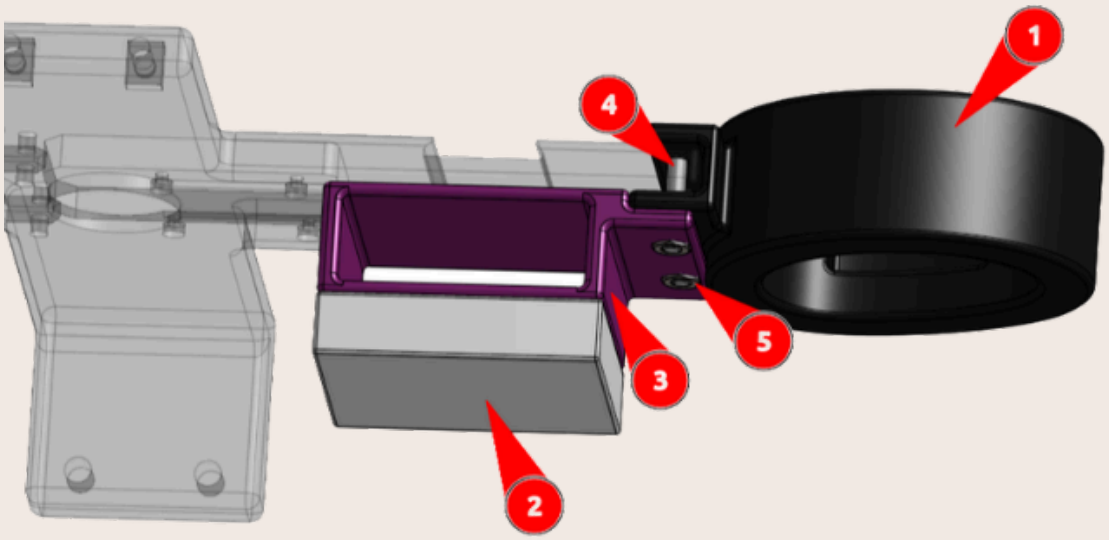


Figure 3. Controller support positioned beneath the main support.

03. Install the Servo-Controller Carrier and Board

Press the left_servo_controller_support into the left_controller_support until it seats in the intended recess. Install the servo-controller board so its connector remains accessible. Confirm that the controller support is fully seated in the slot of handumi-main-support. Connect the servo bus cable, route it away from moving components and fit the servo_controller_cover.

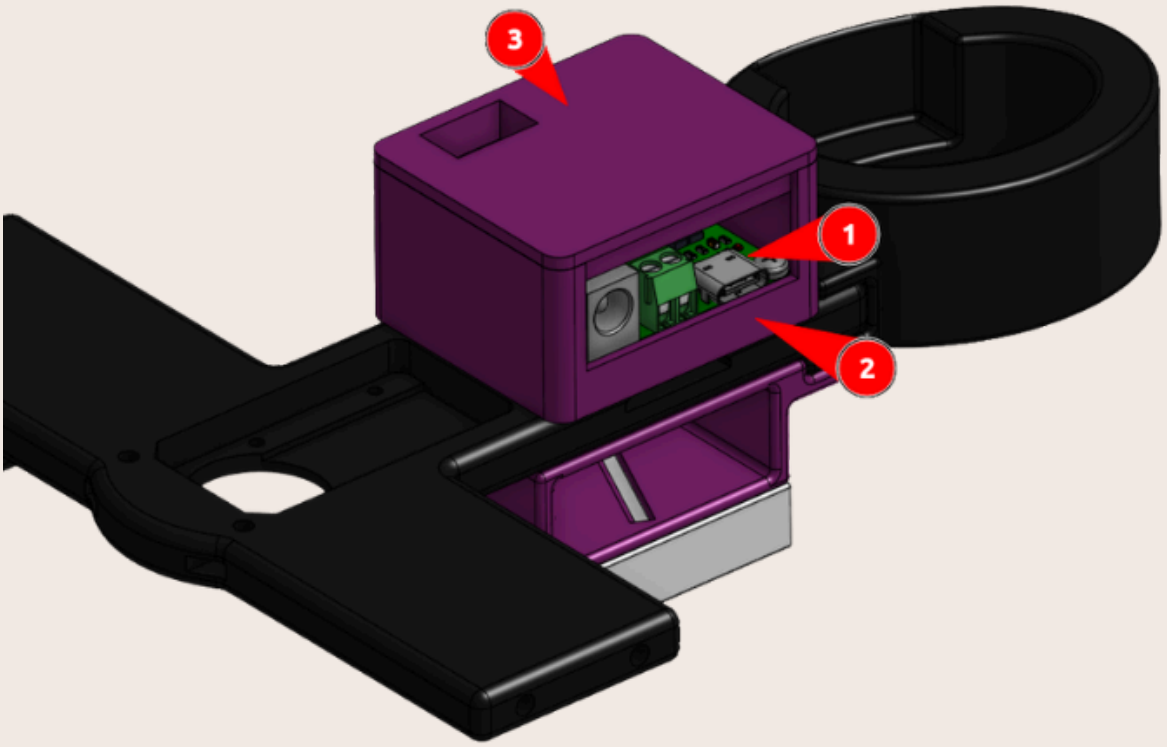


Figure 4. Servo-controller board and accessible connector in the support.

04. Mount the Servo

Seat the servo in its mounting pocket with the output shaft facing the crank mechanism. Secure it with four ANSI B18.6.4 self-tapping screws at the indicated mounting holes. Confirm that the housing is seated and that the output shaft can rotate without interference. See Figures 5 and 6.

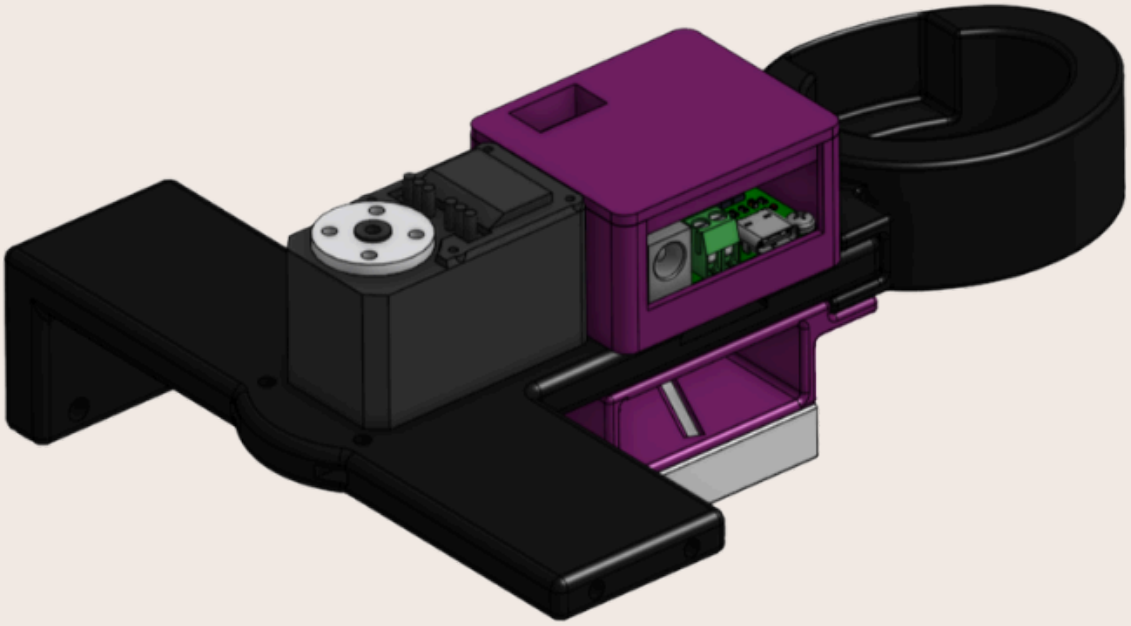


Figure 5. Servo and controller assembly installed on the main support.

05. Inspect the Underside Mounting Interface

Turn the assembly over and confirm that the support, servo and mounting interface match the orientation in Figure 6. Check that no cable or printed edge obstructs the servo output or the future crank plate.

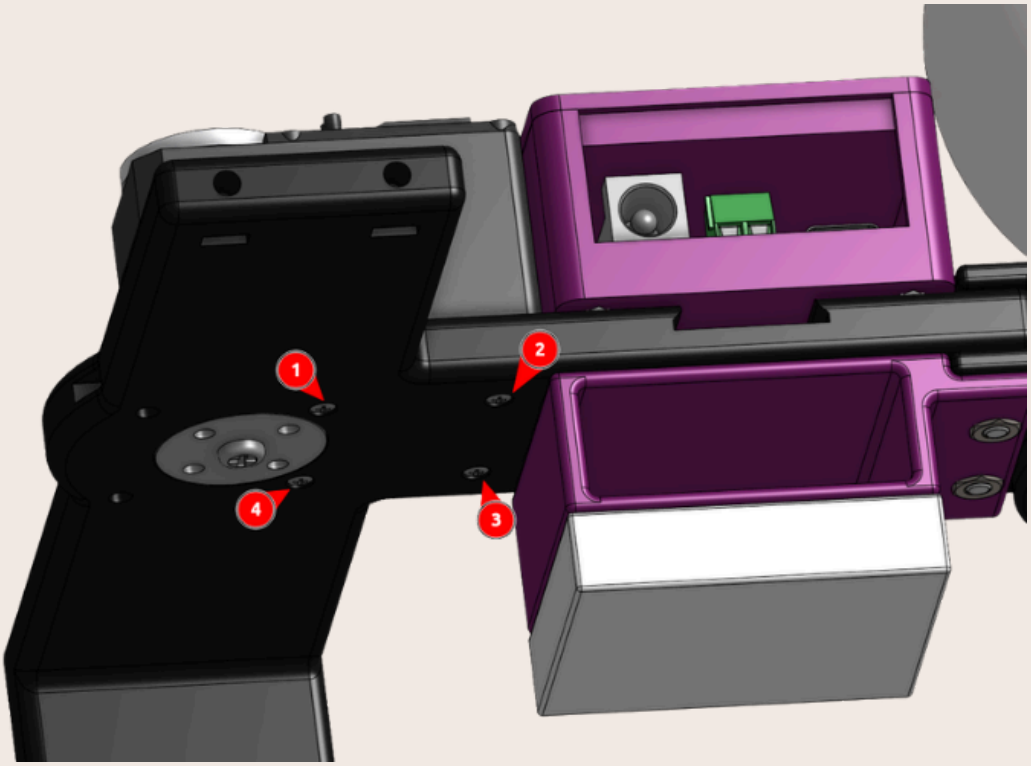


Figure 6. Underside view of the support and servo mounting interface.

06. Assemble the Crank Mechanism

Seat the crank_mechanism_plate on the servo output and align its linkage holes with the two finger-link paths. Secure the plate with four ISO 7045 M3 × 6 screws as specified in the assembly drawing. Install the MR632Z bearings at the locations marked in Figure 7. Rotate the crank by hand and confirm that it clears the surrounding structure.

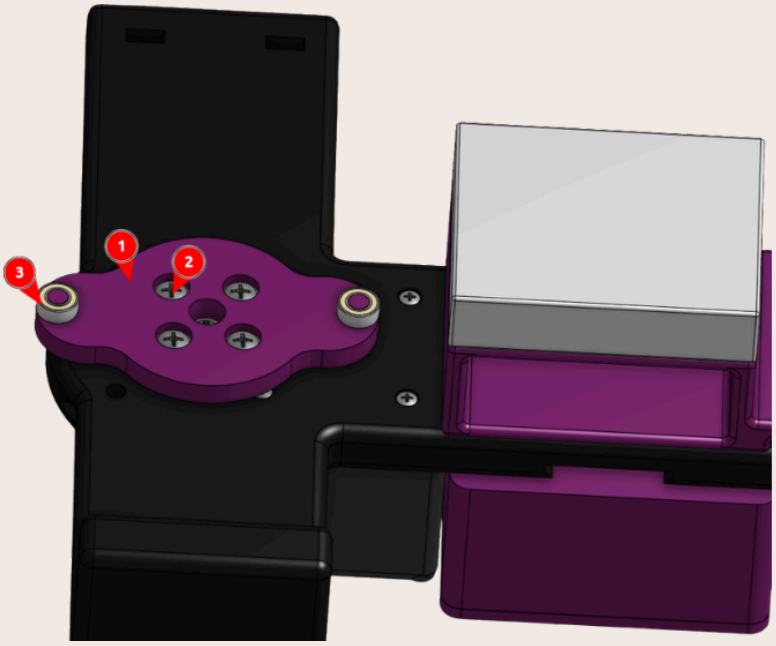


Figure 7. Crank plate and bearing positions beneath the main support.

Press the connecting links and MR632Z bearings into the indicated positions on the crank mechanism, as shown in Figure 8. Seat each bearing squarely and verify that each link pivots freely.

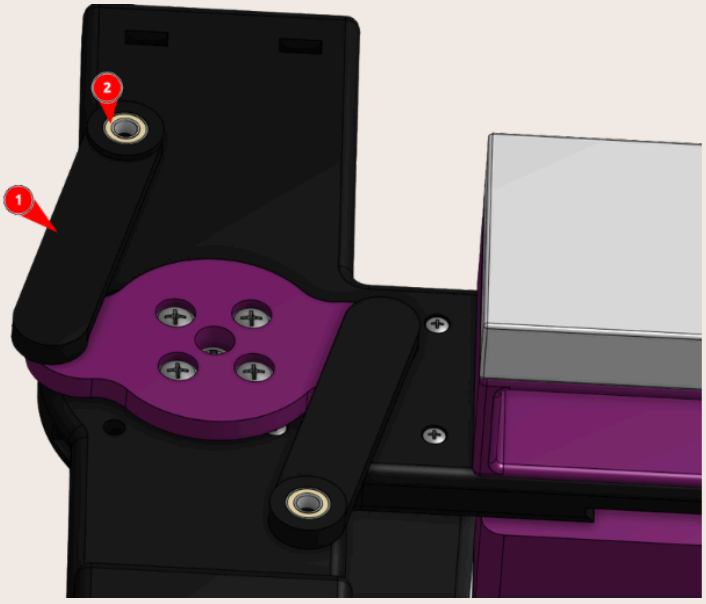


Figure 8. Crank plate with connecting link and bearing installed.

07. Prepare the Finger Links and Grippers

Identify the left_thumb_link and left_index_middle_finger_link by their cradle shapes and mounting ears. Install the LM4 linear bushings and finger rings in their respective seats, as shown in Figure 9. Check that each bushing is fully seated and aligned with its guide-shaft passage.

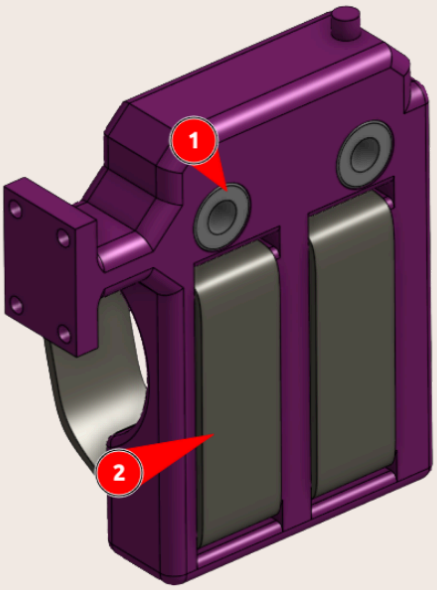


Figure 9. Finger-link bushing seats and finger-ring mounting points.

Attach the matching gripper to each finger link using four DIN 912 M2 × 12 socket-head cap screws and four M2 hex nuts in total, as shown in Figure 10. Keep the grippers paired with their corresponding thumb and index/middle finger links. Confirm that all fasteners are seated and the grippers remain aligned.

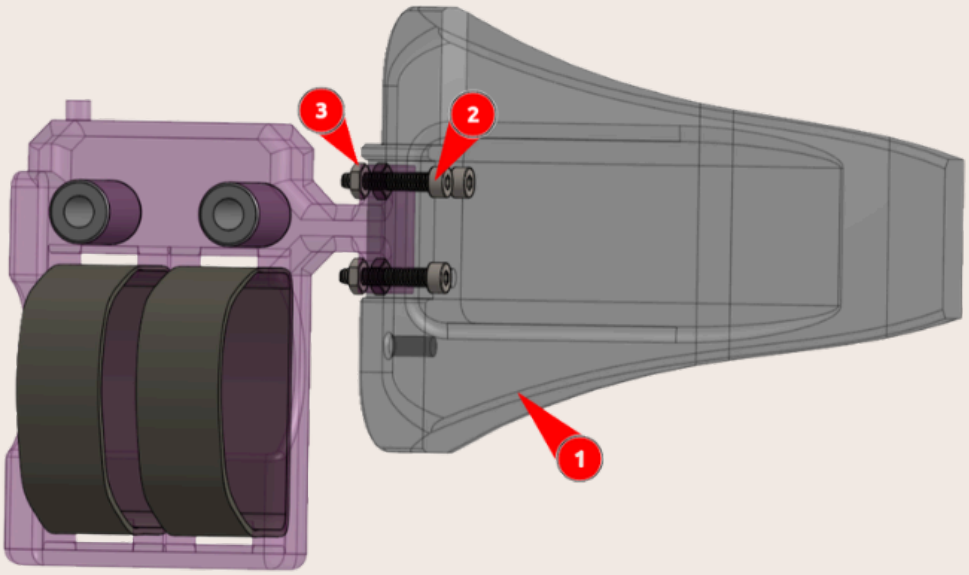


Figure 10. Gripper fasteners and mounting points.

08. Install the Guide Shafts and Finger Links

Insert the two guide shafts through the aligned frame bores and LM4 bushings. Place the thumb and index/middle finger links on their respective sides, keeping the shafts parallel. Press the connecting links onto their corresponding finger-link joints as shown in Figure 11. Move both links through their travel and check for smooth motion without binding.

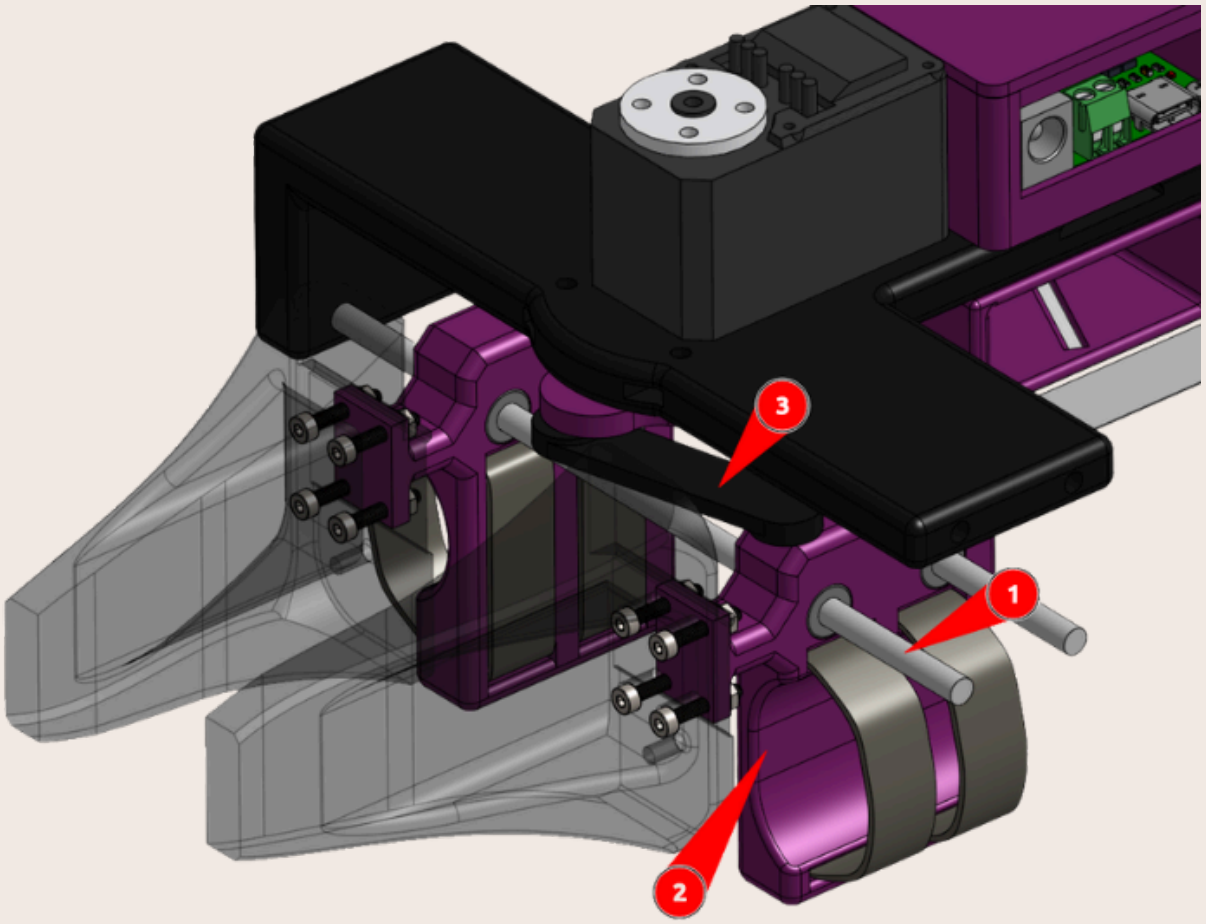


Figure 11. Guide shafts and finger links installed.

09. Fit the Main Support Cover Plate

Align the main_support_cover_plate with the guide shafts and the mounting recess in handumi-main-support. Install two DIN 912 M3 × 15 socket-head cap screws and two M3 hex nuts at the indicated holes. Move the crank through its full travel. Neither connecting link should rub the frame, cross the other link or bind at an end position.

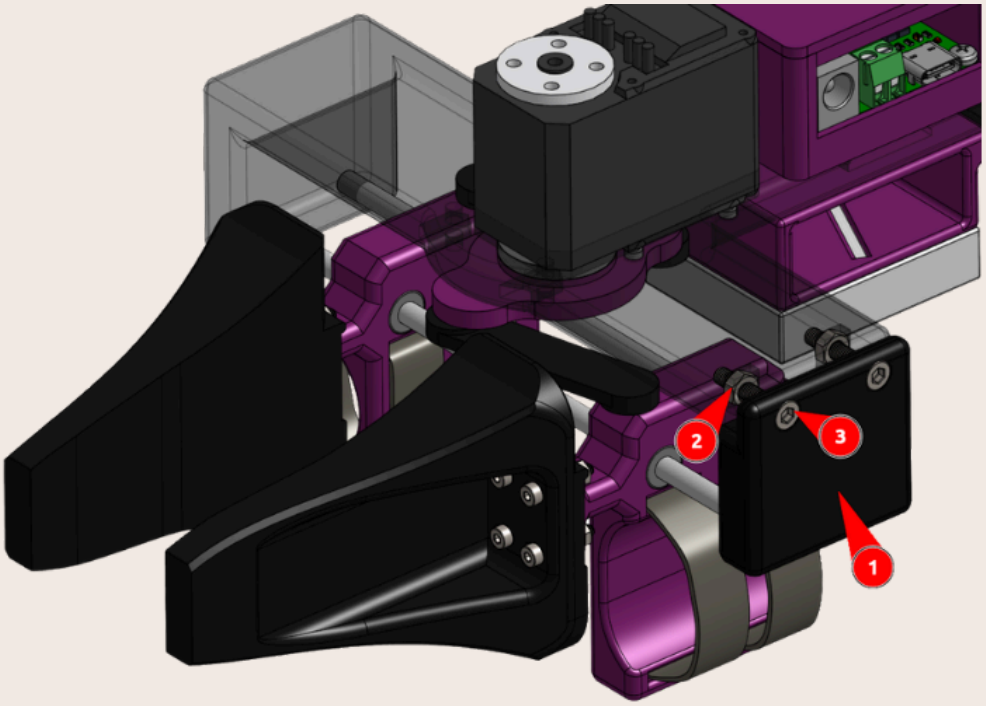


Figure 12. Main support cover plate installed.

10. Install the Camera Support

Position the HandUMI camera support (handumi-camera-support) on the upper forward mounting face of handumi-main-support, with its inclined platform facing the finger working area. Align the holes and secure it with two M2.5 × 8 socket-head screws and two M2.5 hex nuts, as shown in Figure 13. Confirm that the bracket is seated against the frame.

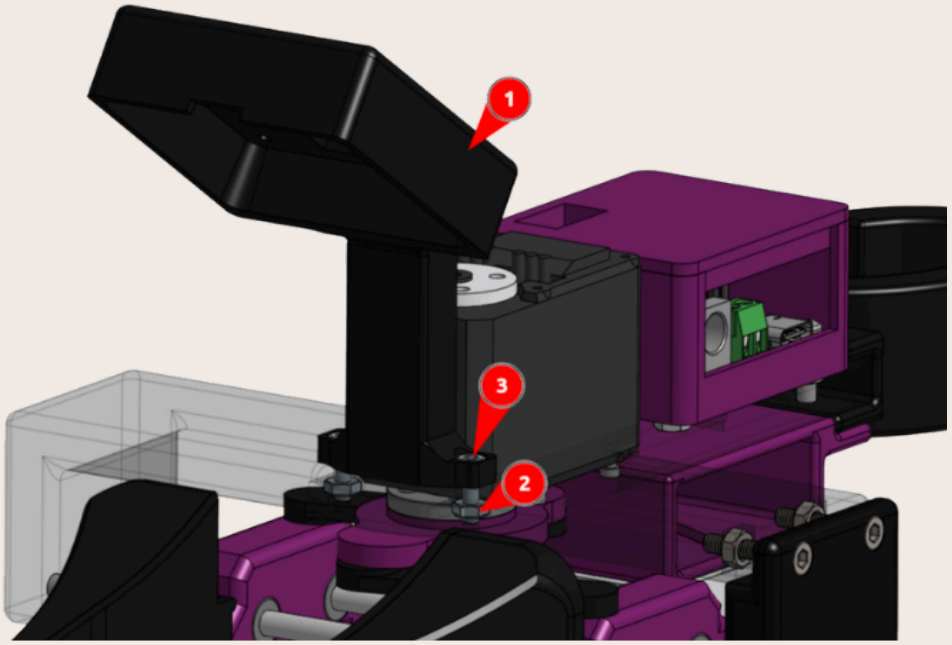


Figure 13. Inclined camera support above the finger mechanism.

11. Mount the Fisheye Camera

Position the fisheye camera on handumi-camera-support with the lens aimed toward the finger working area. Secure it with four DIN 912 M2 × 10 socket-head cap screws and four M2 hex nuts, as shown in Figure 14. Route its cable along fixed structure, clear of the crank and finger links. Check that the camera does not shift when the fingers move.

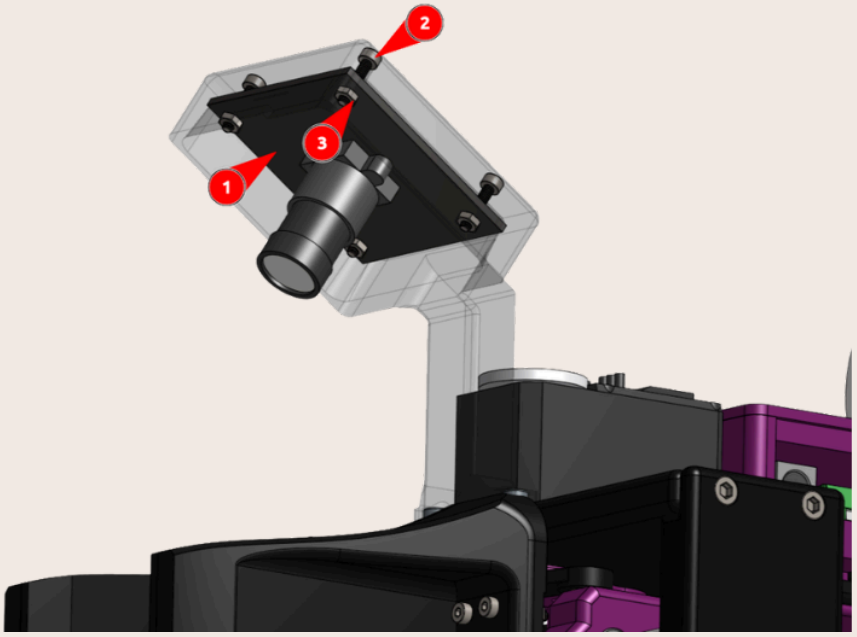


Figure 14. Fisheye camera fasteners and mounting points.

12. Fit the Camera Cover

Press the camera_cover into its mating features until it seats flush. Confirm that the cover does not obstruct the lens or pinch the camera cable. See Figure 15.

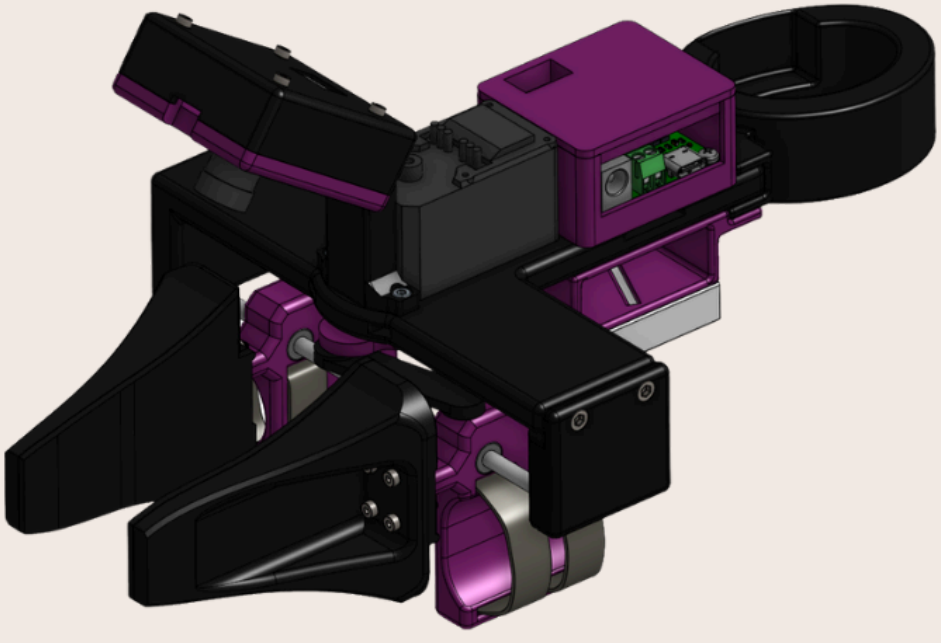


Figure 15. Camera cover installed on the completed assembly.